→ N_s (Sync speed) $= \frac{120f}{P}$ (rpm) (P → no of poles)

$\omega_s = \frac{2\pi N_s}{60}$

stator → AC
rotor → DC

→ $E = 4.44 f \phi T K_\omega$ ↓ generated emf per phase

ϕ → flux per pole
T → turns per phase
α = 180 - coil span angle
β = 180/slots per pole

$K_\omega = K_p K_d$
$K_p = \cos(\alpha/2)$
$K_d = \frac{\sin(m\beta/2)}{m\sin(\beta/2)}$

→ Generator : $E = V + I(R_a + jx_s)$

→ motor : $V = E + I(R_a + jx_s)$

→ Power $P = \frac{EV}{x_s}\sin\delta$ (Real) 3phase power $= \frac{3EV\sin\delta}{x_s}$

δ → power angle

→ $T = \frac{P}{\omega_s} = \frac{3EV\sin\delta}{\omega_s x_s}$

→ $I = \frac{V - E}{R_a + jx_s}$ $\cos\phi = \frac{P}{\sqrt{3}VI}$

→ Reactive Power $Q = \frac{3V(E\cos\delta - V)}{x_s}$

→ $z_s = E/I$ (Synchronus impedance) $z_s^2 = x_s^2 + R_a^2$

→ $V.R = \frac{E - V}{V} \times 100$ → $\delta = \angle E - \angle V$

→ $P_{loss} = P_{cu} + P_{core} + P_{mech}$ $P_{cu} = 3I^2R_a$